

Math 2551 Worksheet: Curves in Space and Their Tangents

1. Describe the graph of the curve $\mathbf{r}(t) = \langle t \cos(t), t \sin(t), t \rangle$, $t \in \mathbb{R}$.
2. Find a vector-valued function for the curve of intersection of the cylinder $x^2 + y^2 = 9$ and the plane $y + z = 2$.
Hint: How could you parameterize the circle $x^2 + y^2 = 9$ in the plane?
3. What is the difference between the parametric curves $\mathbf{f}(t) = \langle t, t, t^2 \rangle$, $\mathbf{g}(t) = \langle t^2, t^2, t^4 \rangle$, and $\mathbf{h}(t) = \langle \sin(t), \sin(t), \sin^2(t) \rangle$ as t runs over all real numbers?
4. With a parametric plot and a set of t values, we can associate a ‘direction’. For example, the curve $\langle \cos(t), \sin(t) \rangle$, $t \in [0, 2\pi]$ is the unit circle traced counterclockwise. How can we change a set of given parametric equations and t values to get the same curve, only traced backwards?
5. The motion of a particle in the xy -plane at time t is described by the vector function

$$\mathbf{r}(t) = e^t \mathbf{i} + \frac{2}{9} e^{2t} \mathbf{j}$$

- (a) Find an equation in x and y whose graph is the path of the particle. Consider how $y(t)$ is related to $x(t)$ and what values $x(t)$ takes on.
 - (b) Find the particle’s velocity and acceleration vectors at $t = \ln(3)$.
 - (c) Sketch the path of the particle and include the particle’s velocity and acceleration vectors at $t = \ln(3)$.
6. Find the parametric equations for the line that is tangent to the curve

$$\vec{r}(t) = \left\langle \ln t, \frac{t-1}{t+2}, t \ln t \right\rangle, \text{ at } t = 1.$$

7. Determine the point at which $\mathbf{f}(t) = \langle t, t^2, t^3 \rangle$ and $\mathbf{g}(t) = \langle \cos(t), \cos(2t), t+1 \rangle$ intersect, and find the angle between the curves at that point. (Hint: You’ll need to set this up like the line intersection problems you’ve seen before, writing one in s and one in t).
If these two functions were the trajectories of two bumblebees on the same scale of time, would the bees collide at their point of intersection? Explain.
8. Find the equation of the plane perpendicular to the curve $\langle \cos(t), \sin(t), \cos(6t) \rangle$ when $t = \pi/4$.